A State-Space Behavioral Model for CMOS Class E Power Amplifiers

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***Abstract*—Class E amplifiers are imposed by time-domain constraints. This makes the design of such amplifiers more difficult than other types of amplifiers. This paper presents a new method to analyze Class E power amplifiers using state-space equations. A description by these equations makes it possible to find the correct initial conditions that leads to the steady-state solution of the amplifier. Furthermore, it is possible to derive the power balance of the power amplifier based on its state-space equations. The presented method has shown to be very valuable when designing switched power amplifiers. Furthermore, the method can also be applied for power amplifiers that do not fulfill the Class E switching conditions.**

***Index Terms*—Class E, CMOS, power amplifier, state-space model.**

I. Introduction

POWER amplifiers are categorized into different classes depending on their behavior. The distinction is made based on their bias points, the working principle of the active device and the voltage or current shaping done by the external passive elements [1]–[3].

When designing a CMOS power amplifier, a Class E working principle [4] is the best choice as these amplifiers are capable of combining a high efficiency (>50%) with a reasonable output power (30 dBm) [5]–[8]. Some main drawbacks appear when these amplifiers are being implemented in a CMOS technology: the high drain–source voltage that occurs may cause breakdown of the switching transistor and a fully integrated solution will degrade the efficiency because of lossy passive elements. Furthermore, nonideal switching due to component mismatch will cause an additional efficiency reduction.

A Class E amplifier is a switching type amplifier. The fact that switching amplifiers are inherently nonlinear shows no obstruction for constant envelope modulation schemes like Global System for Mobile Communications and Bluetooth. Therefore, a Class E amplifier is also a good candidate for fully integrated CMOS transceivers.

The principle of a Class E amplifier is described by time-domain constraints. Hence, a time-domain steady-state solution of the power amplifier is necessary. In order to have a method that is both fast and accurate, a state-space model is used, combined with a standard matrix-based mathematical package. This approach allows one to model all important effects as well as losses of a CMOS implemented Class E amplifier. The method can even be applied to other classes of switching power amplifiers.

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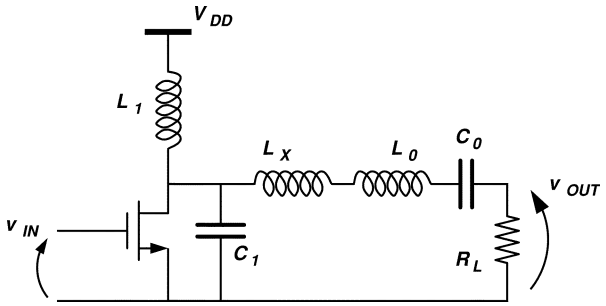


Fig. 1. Basic Class E amplifier topology.

The paper is organized as follows. In the following section, the basics of the Class E amplifier are briefly reviewed. Section III discusses a state-space description of the Class E amplifier. In Section IV, a method will be derived to calculate the power balance of the amplifier by some fast matrix manipulations. The usefulness of the method is demonstrated in Section V and final conclusions are drawn in Section VI.

II. Class E Power Amplifier

A. Working Principle

In order to obtain a high-efficiency power amplifier, two basic ideas must be fulfilled [9]: 1) use the transistor as a switch instead of a current source and 2) minimize all losses that occur when switching the transistor. The first idea is also being used in Class D and F amplifiers and the overdriven versions of Class A and B amplifiers [10]. The second idea is a typical property of Class E amplifiers, where passive components are used to minimize the drain–source voltage when the switch closes [4]. Furthermore, one can demand that the derivative of the switch voltage also equals zero at the switching moment. This demand will make the amplifier less sensitive to timing errors and component variations. Mathematically, the two basic equations describing the Class E operation are given by

$$\text{Class E} \Leftrightarrow \begin{cases} v_{\text{sw}}(t_1) = 0 \\ \left.\dfrac{dv_{\text{sw}}}{dt}\right|_{t=t_1} = 0 \end{cases} \tag{1}$$

where t_1 represents the instant at which the switch closes, and $v_{\text{sw}}(t)$ represents the switch voltage.

Fig. 1 depicts a basic circuit that can satisfy the Class E requirements, provided the correct component values are chosen. If the nMOS transistor is driven by a square wave, it can be seen

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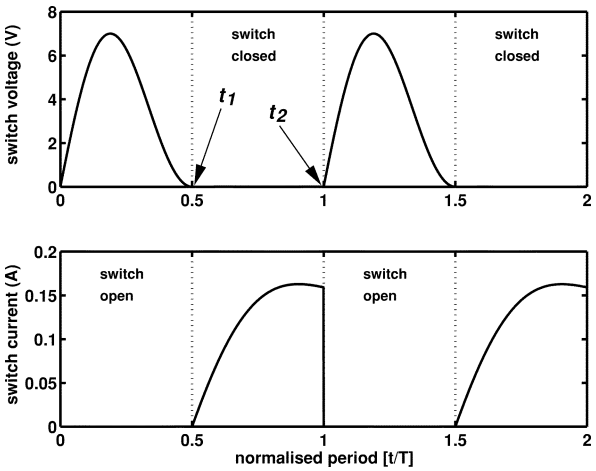


Fig. 2. Switching voltage and current of the Class E amplifier.

as a voltage-controlled switch with a nonzero on-resistance $R_{\rm on}$. In this case, the switch voltage $v_{\rm sw}(t)$ of equation (1) equals the drain–source voltage $v_{DS}(t)$ of the nMOS transistor.

For the Class E amplifier of Fig. 1, the six passive components are constrained by following general considerations. The combination of L_0 and C_0 forms a harmonic filter that is tuned to the switching frequency of the amplifier. The values of L_0 and C_0 will be determined by the switching frequency and the allowed harmonic distortion of the output signal, which is related to the quality factor of this LC filter. The value of the load resistance R_L will mainly be determined by the wanted output power. In a traditional Class E design, L_1 is chosen large so it acts like a current source. Since the Class E working conditions are given by two equations, two components of the circuit can be chosen in such a way that the amplifier fulfills the Class E working conditions. For the circuit of Fig. 1, these two components are the capacitor C_1 and the inductance L_X. In this way, all components are determined provided that equation (1) can be solved. In a properly designed Class E amplifier, the drain–source voltage and drain current will have a shape like in Fig. 2.

B. Solving the Class E Equations

Since the Class E equations are specified in time domain, a time-domain expression for the switch voltage $v_{\rm sw}(t)$ and switch current $i_{\rm sw}(t)$ must be found. Once these expressions are derived, they can be substituted in (1). This will lead to two conditions, determining the values of L_X and C_1. The most difficult task in this approach is to find the expressions $v_{\rm sw}(t)$ and $i_{\rm sw}(t)$. Several methods are known in literature and they can be divided into two main categories.

1) Assumption of Sinusoïdal Output Current: If the quality factor of the output filter Q_L is assumed to be infinite, the output current will have a sinusoïdal shape. Under this assumption, a straightforward analytical expression for $v_{\rm sw}(t)$ and $i_{\rm sw}(t)$ can be found. Raab was the first to derive expressions for the Class E amplifier using this method [11]. He also assumed an infinite value of L_1, a constant supply current I_{DD}. Furthermore, no losses were taken into account which resulted in an efficiency of 100%. Zulinski [12], [13] and Li [14] expanded this method to a finite inductance value and Alinikula [15] followed the same method to take into account the nonlinear junction capacitance of the switch. But all of these methods still assume a high-quality factor for the output filter and a completely lossless circuit. It is clear that these assumptions are no longer valid in a CMOS implementation.

2) Laplace Methods: Another possibility to find an expression for $v_{\rm sw}(t)$ and $i_{\rm sw}(t)$ is by applying the inverse Laplace transformation. This transformation will require the initial conditions of the currents through the coils and the voltages over the capacitors. These initial conditions can be found by solving a set of equations, derived from the steady-state condition. This method has been applied in [16]–[19] and allows one to model the effect of a finite value of L_1 and Q_L. However, if the series resistances of the coils are introduced, it becomes unfeasible to apply an analytical inverse Laplace transformation. A numerical evaluation might still be possible, but the inverse Laplace transformation still has to be evaluated and consumes a lot of CPU time. Examples of this method can be found in [20] and [21]. In these numerical methods, the design of the Class E amplifier is done by using well known optimization techniques.

C. A CMOS Integration of Class E Power Amplifiers

When moving toward a submicron CMOS implementation of a Class E amplifier, several problems will appear [22]. As supply voltage decreases, the currents in the circuit will increase for the same output power and the losses in the RF choke L_1 can no longer be neglected. By lowering the inductance value of this choke, the losses of this choke will also decrease and, as a result, the efficiency of the amplifier will increase. Therefore, it is necessary to take into account the influence of a small value of L_1. Furthermore, a small inductor is easier to integrate in CMOS. The switch consists of an nMOS transistor, incorporating a certain switch resistance $R_{\rm on}$. This resistor will influence the operation of the amplifier and cannot be neglected as efficiency will mainly depend on it. When implemented in a CMOS technology, the output filter L_0C_0 will have a Q_L of about 10. Therefore, the assumption of infinite Q_L no longer holds.

In order to design a CMOS Class E power amplifier, it is desirable to have an analysis and design tool, able to model and take into account all previously mentioned effects. Furthermore, the obtained method should be both fast and accurate so one can explore the design space. The method should also be able to analyze a more general switching power amplifier, not satisfying the Class E working conditions.

III. State-Space Description of the Class E Power Amplifier

A. Formulating the State-Space Equations

Fig. 3 shows the proposed model of a CMOS power amplifier. Notice that all important losses are taken into account. When the transistor is in the off state, it can be modeled as a current source, representing the leakage current $I_{\rm leak}$ which is chosen to be constant. When the switch is on, is can be modeled as a resistor $R_{\rm on}$.

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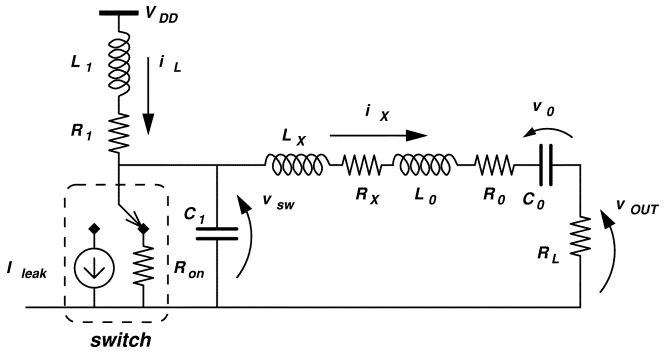


Fig. 3. Class E amplifier model, including all important losses.

The state variables associated with the schematic of Fig. 3 are given by the matrix $\mathbf{q}(t)$

$$\mathbf{q}(t) = \begin{bmatrix} i_L(t) \\ i_x(t) \\ v_{\rm sw}(t) \\ v_0(t) \end{bmatrix}. \tag{2}$$

These variables can also be seen as the energy-storing variables of the circuit. Also, notice that no state degradation occurs.

The state equations can be found by setting up expressions for the first derivative of each state variable. This leads to the following set of equations when the switch is closed:

$$L_1 \frac{di_L(t)}{dt} = V_{DD} - v_{\rm sw}(t) - R_1 i_L(t) \tag{3}$$

$$(L_x + L_0) \frac{di_x(t)}{dt} = v_{\rm sw} - (R_x + R_0 + R_L) i_x(t) - v_0(t) \tag{4}$$

$$C_1 \frac{dv_{\rm sw}(t)}{dt} = i_L(t) - i_x(t) - \frac{v_{\rm sw}(t)}{R_{\rm on}} \tag{5}$$

$$C_0 \frac{dv_0(t)}{dt} = i_x(t). \tag{6}$$

When the switch is in the nonconducting phase, (5) changes to

$$C_1 \frac{dv_{\rm sw}(t)}{dt} = i_L(t) - i_x(t) - I_{\rm leak}. \tag{7}$$

These equations can now be put in the well-known matrix format

$$\frac{d\mathbf{q}(t)}{dt} = \mathbf{A}\mathbf{q}(t) + \mathbf{B}. \tag{8}$$

The matrices $\mathbf{A}$ and $\mathbf{B}$ can be formulated for the two different states of the switch. This leads to two sets of state-space equations of which the first describes the system with the switch open (nonconducting phase) and the second describes the system with a closed switch (conducting phase). So, one obtains the matrices $\mathbf{A_1}$, $\mathbf{B_1}$, $\mathbf{A_2}$, and $\mathbf{B_2}$. Notice that all these matrices are time independent.

B. Solving the State-Space Equations

The general solution of a state-space equation is given by

$$\mathbf{q}(t) = e^{\mathbf{A}t}\mathbf{q_0} + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\, d\tau \tag{9}$$

in which $\mathbf{q_0}$ represents the initial states of the elements of $\mathbf{q}(t)$. Since $\mathbf{A}$ is time independent, the integral in (9) can be written as

$$\int_0^t e^{-\mathbf{A}\tau}\, d\tau = \mathbf{A}^{-1}\left(\mathbf{I} - e^{-\mathbf{A}t}\right). \tag{10}$$

Substituting this expression in (9) leads to

$$\mathbf{q}(t) = e^{\mathbf{A}t}\mathbf{q_0} + e^{\mathbf{A}t}\left(\mathbf{A}^{-1}\left(\mathbf{I} - e^{-\mathbf{A}t}\right)\right)\mathbf{B}. \tag{11}$$

As the matrix product $e^{\mathbf{A}t} \cdot \mathbf{A}^{-1}$ is commutative, this expression can be further simplified to

$$\mathbf{q}(t) = e^{\mathbf{A}t}\mathbf{q_0} + \mathbf{A}^{-1}\left(e^{\mathbf{A}t} - \mathbf{I}\right)\mathbf{B}. \tag{12}$$

The above derivation can be formulated for the two states of the switch. Hence, we obtain

$$\mathbf{q_1}(t) = e^{\mathbf{A_1}t}\mathbf{q_{01}} + \mathbf{A_1}^{-1}\left(e^{\mathbf{A_1}t} - \mathbf{I}\right)\mathbf{B_1} \tag{13}$$

$$\mathbf{q_2}(t) = e^{\mathbf{A_2}t}\mathbf{q_{02}} + \mathbf{A_2}^{-1}\left(e^{\mathbf{A_2}t} - \mathbf{I}\right)\mathbf{B_2}. \tag{14}$$

In order to obtain the solution of the Class E amplifier, it is necessary to find an expression of $\mathbf{q_{01}}$ and $\mathbf{q_{02}}$.

C. Determining the Initial Conditions $\mathbf{q_{01}}$ and $\mathbf{q_{02}}$

As a state variable can always be associated with energy, its value cannot suddenly change. Therefore, the value of the state variables at the end of the conducting phase should be the same as the value of the state variables at the beginning of the nonconducting phase. The same reasoning can be made for the nonconducting phase, leading to the following set of equations:

$$\begin{cases} \mathbf{q_{02}} \equiv \mathbf{q_1}(t_1) = e^{\mathbf{A_1}t_1}\mathbf{q_{01}} + \mathbf{A_1}^{-1}\left(e^{\mathbf{A_1}t_1} - \mathbf{I}\right)\mathbf{B_1} \\ \mathbf{q_{01}} \equiv \mathbf{q_2}(t_2) = e^{\mathbf{A_2}t_2}\mathbf{q_{02}} + \mathbf{A_2}^{-1}\left(e^{\mathbf{A_2}t_2} - \mathbf{I}\right)\mathbf{B_2} \end{cases} \tag{15}$$

which can easily be solved toward $\mathbf{q_{01}}$ and $\mathbf{q_{02}}$. In these equations, t_1 is the instant at which the switch closes and is equal to δT, with δ the duty cycle and T the period of the input signal. Note that both solutions are formulated, starting from $t = 0$. Therefore, the instant the switch enters the nonconducting phase should be defined as $t_2 = (1 - \delta)T$.

A final difficulty lies in the evaluation $e^{\mathbf{A}t}$. However, this expression only has to be evaluated at two instants t_1 and t_2. Evaluating such a matrix can easily be done numerically (e.g., with MATLAB) or by finding the eigenvalues of $\mathbf{A}$. It should be noted that (15) forms a linear set of equations which is easy to solve. Furthermore, evaluating $e^{\mathbf{A}t}$ and solving a linear set of two equations will be performed much faster than solving the set of differential equations given by (3)–(6). Once the steady-state initial conditions $\mathbf{q_{01}}$ and $\mathbf{q_{02}}$ are known, then they can be substituted in expressions (13) and (14). The power amplifier can now be analyzed very fast. Furthermore, this approach clearly allows one not only to evaluate Class E power amplifiers, but deviations from the Class E conditions can be analyzed as well.

Analyzing a power amplifier mainly comprises calculating the output power and the efficiency. Therefore, a method to analyze these numbers, based on the state-space description is also presented.

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D. Designing for Class E Conditions

In order to have a Class E power amplifier, both the switching voltage and its first derivative should be zero at the beginning of the conducting phase. Since the value of $v_{\rm sw}(t_1)$ is incorporated in $\mathbf{q_{02}}$, this requires no further calculation. The derivative of $\mathbf{q_{02}}$ can be found by applying (8), and only requiring a matrix multiplication and an addition.

By changing the values of L_X and C_1, the value of $v_{\rm sw}(t_1)$ and $dv_{\rm sw}(t_1)/dt$ can be minimized. This minimization process can be implemented in various ways. A standard least square error technique was used to combine both stability and speed. The method will vary the values of L_X and C_1 until a satisfactory solution is found. A starting point for this optimization technique is provided by the simple formulas of [11].

IV. Calculating Power and Efficiency

A. General Power Expressions

The output power of the circuit of Fig. 3 is given by

$$P_{\rm out} = R_L \cdot \frac{1}{T} \cdot \int_0^T i_x^2(t)\, dt. \tag{16}$$

By defining the following matrices,

$$\mathbf{W_1} = \int_0^{t_1} \mathbf{q_1 q_1}^{\mathbf{T}}\, dt$$

$$\mathbf{W_2} = \int_0^{t_2} \mathbf{q_2 q_2}^{\mathbf{T}}\, dt$$

the output power can easily be written as

$$P_{\rm out} = \frac{R_L}{T} \cdot \{\mathbf{W_1}[2,\,2] + \mathbf{W_2}[2,\,2]\}\,. \tag{17}$$

To evaluate the power dissipation in the coils L_1, L_X, and L_0, one can apply a similar method.

The dc input power of the Class E amplifier can be written as

$$P_{\rm DC} = \frac{1}{T}\,\mathbf{C}_{\rm DC} \left(\int_0^{t_1} \mathbf{q_1}\, dt + \int_0^{t_2} \mathbf{q_2}\, dt \right) \mathbf{C}_{\rm DC}^{\mathbf{T}} \tag{18}$$

in which

$$\mathbf{C}_{\rm DC} = \left[\sqrt{V_{DD}} \;\; 0 \;\; 0 \;\; 0 \right].$$

The power loss in the switch can be divided in two parts. The power dissipated during the conducting phase is given by

$$P_{\rm sw,\,on} = \frac{1}{T}\,\mathbf{C}_{\rm sw,\,on} \left(\int_0^{t_2} \mathbf{q_2 q_2^T}\, dt \right) \mathbf{C}_{\rm sw,\,on}^{\mathbf{T}} \tag{19}$$

with

$$\mathbf{C}_{\rm sw,\,on} = \left[0 \quad 0 \quad \frac{1}{\sqrt{R_{\rm on}}} \quad 0 \right]$$

which can also be expressed as

$$P_{\rm sw,\,on} = \frac{1}{R_{\rm on}} \cdot \frac{1}{T} \cdot \mathbf{W_2}[3,\,3]. \tag{20}$$

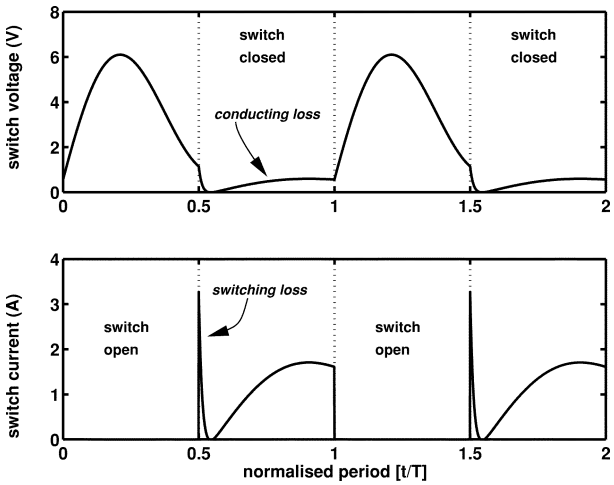


Fig. 4. Switching voltage and current of a nonideal Class E amplifier.

During the nonconducting phase, the power dissipated by the leakage current is given by

$$P_{\rm sw,\,off} = \frac{1}{T}\,\mathbf{C}_{\rm sw,\,off} \left(\int_0^{t_1} \mathbf{q_1}\, dt \right) \mathbf{C}_{\rm sw,\,off}^{\mathbf{T}} \tag{21}$$

in which

$$\mathbf{C}_{\rm sw,\,off} = [\,0 \quad 0 \quad \sqrt{I_{\rm leak}} \quad 0\,].$$

The total power dissipated by the switching transistor is the sum of $P_{\rm sw,\,on}$ and $P_{\rm sw,\,off}$. Fig. 4 show the waveforms when nonideal switching occurs. The loss $P_{\rm sw,\,on}$ can be divided in a nonideal switching loss and a conducting loss.

All the above equations need a value or expression for $\int_0^T \mathbf{q}\, dt$ and $\int_0^T \mathbf{qq^T}\, dt$. This forms the topic of the following part.

B. Determining $\int_0^T \mathbf{q}\, dt$ and $\int_0^T \mathbf{qq^T}\, dt$

Integrating the basic state-space equation (8) and considering the fact that $\mathbf{A}$ and $\mathbf{B}$ are time independent leads to

$$\int_0^T \mathbf{q}\, dt = \mathbf{A}^{-1}(\mathbf{q}(T) - \mathbf{q}(0) - \mathbf{B}T) \tag{22}$$

and the equivalent expression

$$\int_0^T \mathbf{q^T}\, dt = \left(\mathbf{q^T}(T) - \mathbf{q^T}(0) - \mathbf{B^T}T\right) \left(\mathbf{A^T}\right)^{-1}. \tag{23}$$

The derivation of $\int_0^T \mathbf{qq^T\,dt}$ is somewhat more cumbersome. It starts by taking the following derivative:

$$\begin{aligned} \frac{d\mathbf{qq^T}}{dt} &= \frac{d\mathbf{q}}{dt}\,\mathbf{q^T} + \mathbf{q}\frac{\mathbf{dq^T}}{\mathbf{dt}} \\ &= \mathbf{Aqq^T} + \mathbf{Bq^T} + \mathbf{qq^T A^T} + \mathbf{qB^T}. \end{aligned} \tag{24}$$

Integrating this expression leads to

$$\begin{aligned} \mathbf{qq^T}\big|_0^T &= \mathbf{qq^T}(T) - \mathbf{qq^T}(0) \\ &= \mathbf{A} \int_0^T \mathbf{qq^T}\, dt + \mathbf{B} \int_0^T \mathbf{q^T}\, dt \\ &\quad + \int_0^T \mathbf{qq^T}\, dt\mathbf{A^T} + \int_0^T \mathbf{q}\, dt\mathbf{B^T}. \end{aligned} \tag{25}$$

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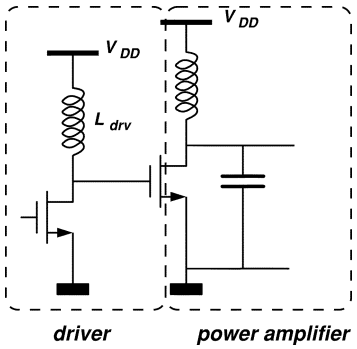


Fig. 5. Typical driver stage for a CMOS Class E power amplifier.

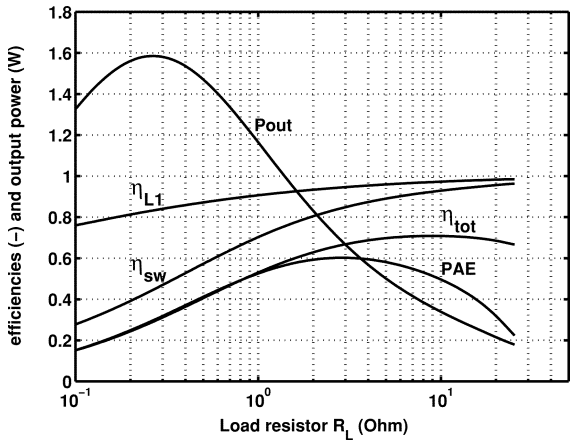


Fig. 6. Variation of output power and PAE as a function of the load resistor.

By using expressions (22) and (23), the above equation can be simplified to

$$\mathbf{AW} + \mathbf{WA}^{\mathbf{T}} + \mathbf{G} = \mathbf{0} \tag{26}$$

in which

$$\begin{aligned}\mathbf{G} = \mathbf{A}^{-1}&(\mathbf{q}(T) - \mathbf{q}(0) - \mathbf{B}T)\mathbf{B}^{\mathbf{T}} \\ &+ \mathbf{B}\left(\mathbf{q}^{\mathbf{T}}(T) - \mathbf{q}^{\mathbf{T}}(0) - \mathbf{B}^{\mathbf{T}}T\right)\left(\mathbf{A}^{\mathbf{T}}\right)^{-1} \\ &- \left(\mathbf{q}\mathbf{q}^{\mathbf{T}}(T) - \mathbf{q}\mathbf{q}^{\mathbf{T}}(0)\right).\end{aligned}$$

Equation (26) is the Lyapunov equation [23], and can be solved for the unknown variable $\mathbf{W}$ since the matrices $\mathbf{A}$ and $\mathbf{G}$ are completely determined. The computation of $\mathbf{W}$ can be done e.g., in MATLAB by standard matrix techniques.

C. Drain Efficiency and Power Added Efficiency

After calculating $\mathbf{W}$ by solving the Lyapunov equation, it can be substituted in the expressions of each of the calculated power. Once the dc input power, the output power, and the different power losses are calculated, one can evaluate the efficiency of the power amplifier. This efficiency η can be seen as the product of the intermediate efficiencies [8]

$$\eta = \frac{P_{\text{out}}}{P_{\text{DC}}} = \eta_{L_1} \cdot \eta_{\text{switch}} \cdot \eta_{L_X} \cdot \eta_{L_0}. \tag{27}$$

For example, the efficiency of the switch is given by

$$\eta_{\text{switch}} = \left(\frac{P_{\text{in}} - P_{R1} - P_{\text{switch}}}{P_{\text{in}} - P_{R1}}\right) = \left(1 - \frac{P_{\text{switch}}}{\eta_{L_1} P_{\text{in}}}\right). \tag{28}$$

The Power Added Efficiency (PAE) takes into account the power needed to drive the large switching transistor. The definition is given by (29), in which P_{driver} is equal to the power, dissipated in the driver stage

$$\mathbf{PAE} = \frac{P_{\text{out}} - P_{\text{driver}}}{P_{\text{DC}}}. \tag{29}$$

In most CMOS implementations, the driver circuit looks like Fig. 5. The driver coil L_{drv} is used to tune out the gate capacitance of the power transistor. This circuit shows some equivalence with the basic Class E circuit of Fig. 1. In fact, the same method derived above was used by the authors to calculate the power needed to drive the switching transistor. If the switching transistor is directly driven by a tapered buffer stage, the driver power is dependent on the gate capacitance of the switching transistor and the used tapering factor [24]–[26].

V. Evaluating the State-Space Approach

The described method was implemented as a set of MATLAB functions. The complete design and evaluation of a Class E amplifier only takes 0.7 s on an HP-UX 11 B2600 system, slightly depending on the initial guess. Therefore, the state-space method can be used to generate some useful tradeoff curves in a reasonable amount of time. An example of the different design options in a CMOS Class E power amplifier will be discussed next, clearly showing the influence of the effects described in Section II-C.

Since the Class E conditions will impose the value of C_1 and L_X, only the load resistor R_L, the supply coil L_1, and the transistor size W/L can be chosen freely. The amplifier is designed at 1 GHz in a standard 0.35-μm CMOS technology. The inductors are made of bonding wires, and their losses are typically about 0.1 Ω/nH.

The 0.35-μm CMOS technology has a breakdown voltage of 7 V. The voltage at the drain of the switching transistor will reach a value of approximately 3.56 times the supply voltage. This number was derived by Raab [11] for the ideal case, but will be different when all the losses are taken into account. The supply voltage of the power amplifier, hence, should be lowered to an appropriate value as to avoid breakdown of the switching transistor. This can easily be implemented by determining the maximum value of the drain voltage and comparing it with the breakdown voltage. As the supply voltage of the power amplifier changes, also the dc input power of the amplifier will change as the matrix $\mathbf{C}_{\text{DC}}$ depends on V_{DD}.

A. Variation of R_L

Fig. 6 shows the value of P_{out} and the efficiencies, for different values of R_L. Each point of the curve represents another Class E power amplifier. As discussed previously, the supply voltage is adapted in every point to prevent breakdown of theswitching transistor. The supply coil L_1 is chosen to be 2 nH and the switch transistor has a width of 6000 μm.

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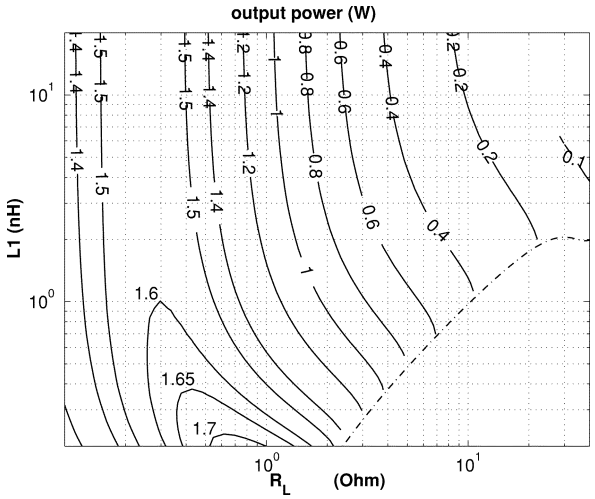


Fig. 7. Contourplot of the output power as a function of R_L and L_1.

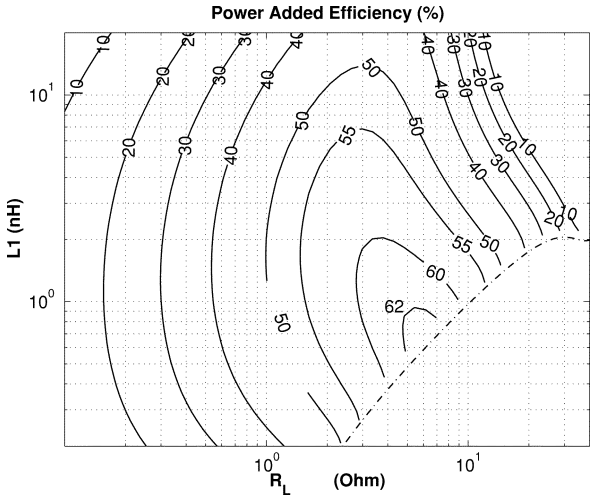


Fig. 8. Contourplot of the PAE as a function of R_L and L_1.

The figure clearly shows that the design can be aimed for either maximal output power ($R_L = 0.25\ \Omega$) or maximal PAE ($R_L = 3\ \Omega$). Hence, choosing a value of R_L smaller than $0.25\,\Omega$ or larger than $3\ \Omega$ has no benefit in this design. Lowering the load resistance R_L will increase the switch current and decrease the efficiency of the transistor $\eta_{\rm sw}$ and this will also decrease the PAE and the output power. For higher values of R_L, the low value of $P_{\rm out}$ will decrease the PAE, but the efficiency of the amplifier $\eta_{\rm tot}$ is high. Notice that $\eta_{\rm tot}$ is dominated by the switch resistor. Also interesting is the maximal output power that can be achieved with the chosen values for W and L_1.

B. Variation of L_1

The influence of L_1 is shown in Figs. 7 and 8. When this inductor is reduced, the value of L_X will decrease toward zero. From a certain point on, the minimal value of L_1 to have Class E conditions is reached. This borderline is indicated by a dashed line on the figures and clearly limits the design space at small values of L_1. The important conclusion of these figures is that a smaller value of L_1 will lead to a larger output power for the same value of R_L. Furthermore, lowering L_1 until L_X becomes zero, always results in an optimal point in terms of output power and PAE.

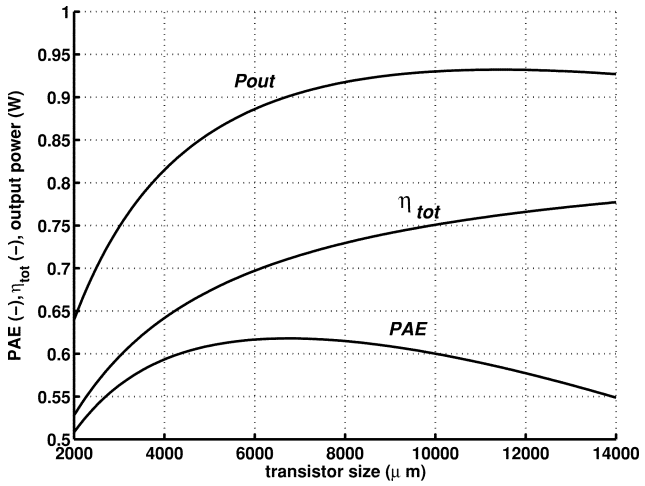


Fig. 9. Variation of output power, efficiency and the PAE as a function of the transistor size. $L_1 = 2$ nH and $R_L = 2\ \Omega$.

C. Variation of W/L

The proposed method allows one to easily investigate the influence of the W/L of the switching transistor. A larger transistor size will lower the on-resistance, but it will require more power from the driver stage and consume more silicon area. Hence, an optimum can be expected for the PAE. This is shown in Fig. 9, where the size of the switching transistor is varied from 2000 to 14 000 μm. Notice that a very large value of W/L is beneficial for the efficiency and the output power, but it will almost be impossible to drive this transistor, resulting in a low PAE. Furthermore, a 14 000-μm-wide transistor is not a practical value for a fully integrated CMOS solution.

All the previous curves lead to an optimal choice of passive components and switch size. A comparison between the results of the presented tool and some published CMOS amplifiers is given in [27].

VI. Conclusion

A method was presented to analyze a CMOS Class E power amplifier. To gain insight in the different tradeoffs, all the losses of the amplifier have to be taken into account and an analytical solution, therefore, becomes unfeasible. To solve the differential equations of the amplifier, a state-space method was proposed, capable to find the steady-state solution in a reasonable time.

The method was implemented as a set of MatLab functions and allows a fast evaluation and an accurate design of CMOS class E power amplifiers, taking into account all important effects of a fully integrated solution. A fast design method allows one to explore the design space and to visualize the different tradeoffs.

This exploration has lead to the conclusion that lowering de DC inductor L_1 is beneficial both for output power and power-added efficiency. Choosing a larger switch transistor is beneficial for the output stage, but not for the driver stage, leading to an optimal value of the transistor size.

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